

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

MODULE - 1

**SATELLITES AND THEIR IMPORTANCE AND
TYPES OF ORBITS - 1**

SATELLITE

- A satellite is a planet, moon or a machine that orbits a star or a planet.
- For example, Earth is a satellite because it orbits the sun. Similarly, the moon is a satellite because it orbits Earth.
- Earth and the moon are examples of natural satellites.
- Thousands of artificial, or man-made, satellites orbit Earth.
- Satellites are used for many purposes like Earth imaging, communication, remote sensing etc.



WHY ARE SATELLITES IMPORTANT?

- 1** The bird's-eye view that satellites have allows them to see large areas of Earth at one time. This ability means satellites can collect more data, more quickly, than instruments on the ground.
- 2** Satellite Telescopes (Ex: Hubble Space Telescope) can see into space better than telescopes at Earth's surface. That's because satellites fly above the clouds, dust and molecules in the atmosphere that can block the view from ground level.
- 3** Satellites are very important for communication. With satellites, TV and phone signals are sent upward to a satellite. Then, almost instantly, the satellite can send them back down to different locations on Earth.



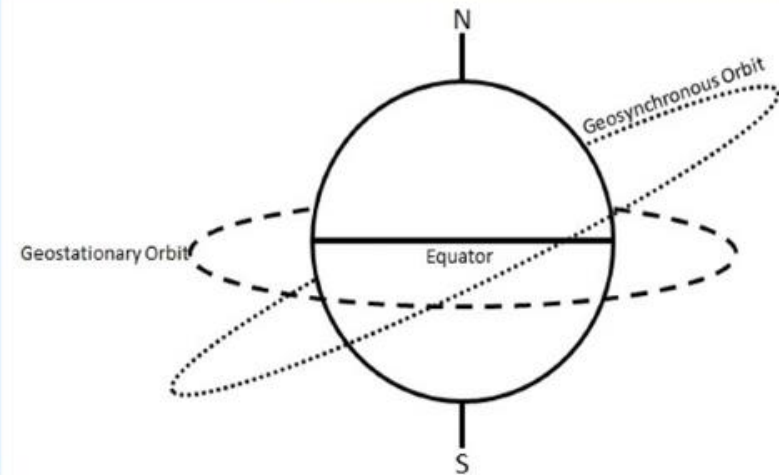
HOW DO SATELLITES ORBIT EARTH?

- Satellites are launched into space on rockets.
- A satellite orbits Earth when its speed is balanced by the pull of Earth's gravity. Without this balance, the satellite would fly in a straight line off into space or fall back to Earth.
- Satellites orbit Earth at different heights, different speeds and along different paths.



GEOSYNCHRONOUS ORBIT

- Also known as the High Earth Orbit.
 - When a satellite reaches 35,786 km from the Earth's surface, it enters a place where the satellite orbits at the same speed that the Earth is rotating.
 - It means the orbital period of satellite is equal to 23 hours 56 minutes and 4 seconds which is the time taken by the Earth to finish one rotation around its own axis.
-
- For an observer on Earth's surface a satellite in this orbit returns to exactly the same position in the sky after a period of one day (23 hours 56 minutes and 4 seconds). It means there is synchronization between the rotation of the Earth and the orbital period of the satellite in geosynchronous orbit.
 - Geosynchronous orbit is an inclined orbit.

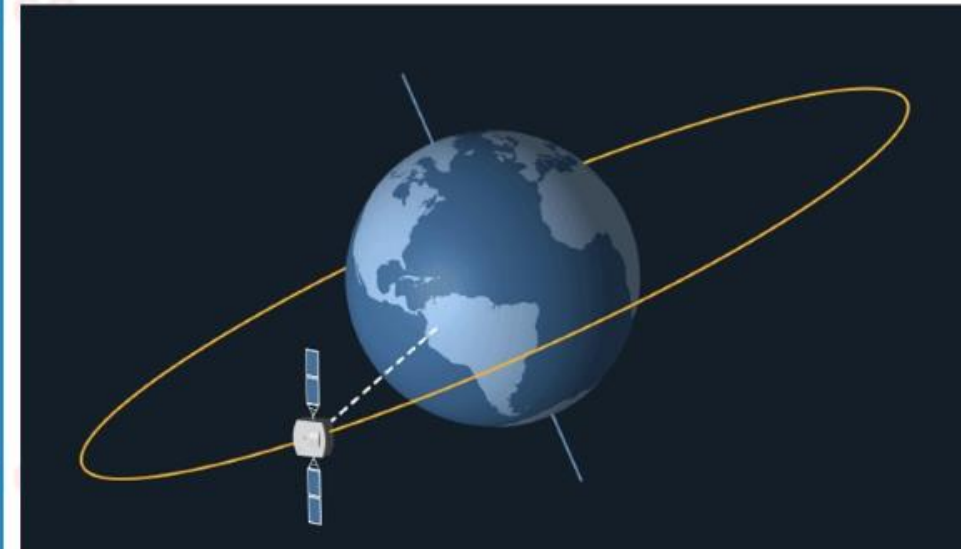


Orbital Period:

The orbital period is the time taken by a satellite to complete one revolution in an orbit around a planet or a star.

GEOSTATIONARY ORBIT

- It is a type of Geosynchronous orbit.
- It is a circular orbit at a height of 35,786 kms above the Earth's equator. A satellite in this orbit follows the direction of the Earth's rotation (from West to East) and has orbital period equal to the Earth's rotational period.
- So, the satellite appears to be motionless and fixed at a position in the sky from ground.
- Some communication satellites and weather monitoring satellites are placed in this orbit so that the antennas at the ground that communicate with the satellite do not have to rotate to keep in constant touch with it.
- The antennas can be pointed permanently at the position in the sky where the satellites are located.



GEOSTATIONARY ORBIT

- India's INSAT series of satellites are placed in this orbit.
- INSAT- Indian National Satellite System: are a series of 11 satellites (24 were launched out of which 11 satellites are still in operation). The 1st INSAT satellite, INSAT 1B was launched in 1983
- INSAT satellites are used for telecommunications, broadcasting, meteorology and search & rescue operations.
- INSAT program is a joint venture of the Department of Space, Department of Telecommunications, Indian Meteorological Department, All India Radio and Doordarshan.
- INSAT system brought a revolution in India's television and radio broadcasting, telecommunications and weather forecasting.



INSAT - 1B



WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

info.learningspacedigital@gmail.com

Visit us at:

www.learningspacedigital.com

0866 244 44 72

98499 42299

94415 27588